

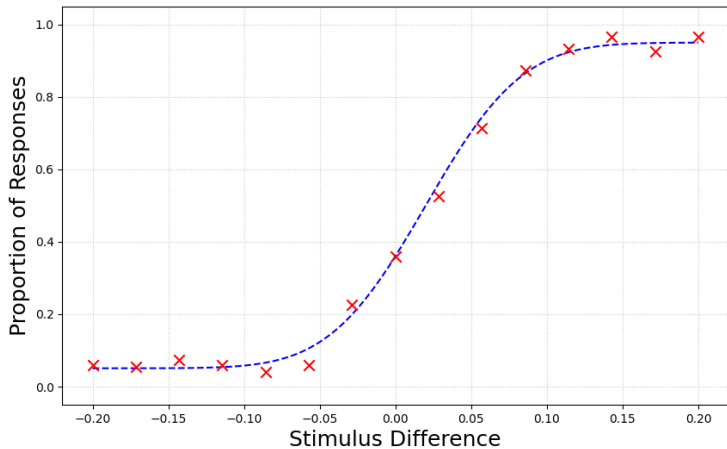
A costly-information foundation for psychometric curve

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Motivation

- ▶ Perception comes before action
 - ▶ firm fundamentals → investment, candidate quality → recruitment, symptom → prescription, etc
- ▶ Psychophysics studies how we perceive
 - ▶ use perceptual tasks, e.g., brightness, pitch, weight, motion
 - ▶ summarize data by psychometric function
- ▶ Sigmoidal curves are widely observed
 - ▶ Fechner (1860)' law: subjective sensation is proportional to the logarithm of the stimulus intensity
 - ▶ Recent explanation: Signal detection theory, Drift diffusion model, and Efficient coding
- ▶ Micro foundation: optimal trade-off between benefit and cost of information processing



Rational Inattention Framework

Decision maker

- ▶ faces state-contingent payoffs and uncertainty about the state
 - ▶ model stimulus strength as the state
- ▶ chooses a signal structure with a certain cost
 - ▶ measured by Fisher information cost (Hébert and Woodford, 2021)
- ▶ takes action optimally according to the received signal
 - ▶ equivalently, state-dependent choice rules/response curve
- ▶ chooses optimally to maximize the gain minus cost

Main result: under uniform prior and increasing relative payoff, any (interior) optimal response curve is sigmoidal.

Fisher information cost

- ▶ Fisher information captures local distinguishability,

$$\mathcal{I}_F(\{\sigma_\theta\}) := E_s[(\frac{\partial}{\partial\theta} \log \sigma(s|\theta))^2]$$

- ▶ Fisher information cost measures overall responsiveness

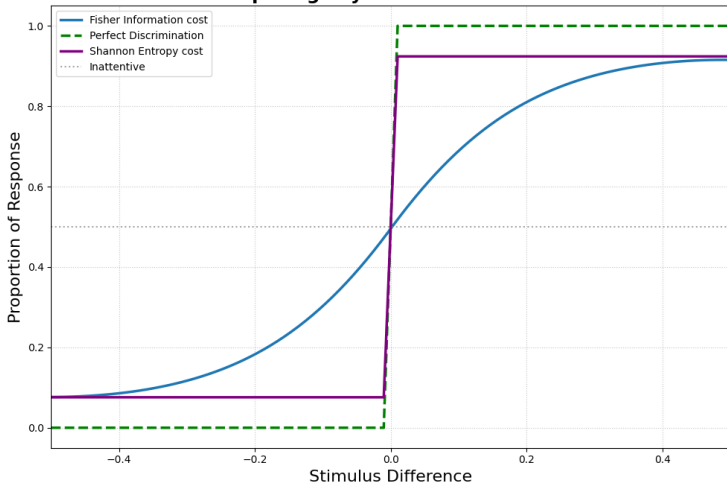
$$C_{FI}(\sigma) := E_{\pi(\theta)}[\mathcal{I}_F(\{\sigma(\theta)\})] = \int_{\Theta} \pi(\theta) \sum_{s \in \mathcal{S}} \frac{(\frac{\partial}{\partial\theta} \sigma(s|\theta))^2}{\sigma(s|\theta)} d\theta.$$

- ▶ Intuition: Penalizes sharp distinctions between nearby states, especially those near 0/1

Why Fisher information cost?

- ▶ Shannon model fails to capture perceptual distance and overestimates DM's response to incentives (Dean and Neligh, 2023)
- ▶ Fisher information cost can be considered as continuous extension of *Neighborhood-based* cost (Hébert and Woodford, 2021), *Log-Likelihood* cost (Pomatto et al., 2023), *Total Information* cost (Bloedel and Zhong, 2024), *f-Information* cost (Bloedel et al., 2025)
 - ▶ Fisher information is the 2nd order gradient of f-divergence

Comparing Psychometric Functions



A binary choice problem (2AFC)

- ▶ State space: Θ , (full-support) Prior: $\rho \in \Delta(\Theta)$
- ▶ Action Set: $A = \{L, R\}$
- ▶ State-contingent payoff: $u(t, L) \equiv 0$, denote $u(\theta, R) := u(\theta)$
- ▶ Optimal signal is binary: $p(\theta) := \sigma(R|\theta)$ for the probability of getting a recommendation of taking R under state θ
- ▶ Expected gain :

$$\int_{\Theta} \rho(\theta) u(\theta) p(\theta) d\theta$$

- ▶ Cost of information:

$$C_{FI}(p, \pi) = \int_{\Theta} \rho(t) \frac{(p'(\theta))^2}{p(\theta)(1-p(\theta))} d\theta,$$

we treat

$$\frac{(p'(\theta))^2}{p(\theta)(1-p(\theta))} = \begin{cases} \frac{(p'(\theta))^2}{p(\theta)(1-p(\theta))}, & \text{if } p(\theta) \in (0, 1) \\ 0, & \text{if } p'(\theta) = 0 \text{ and } p(\theta) = 0 \text{ or } 1. \\ +\infty & \text{other cases} \end{cases}$$

The Optimization Problem

DM chooses a state-contingent response rule $p(\theta)$:

$$\begin{aligned} & \underset{p \in AC(\Theta, [0,1])}{\text{maximize}} \int_{\Theta} \rho(\theta) u(\theta) p(\theta) d\theta - \lambda C_{FI}(p, \rho) \\ \Leftrightarrow & \underset{p \in AC(\Theta, [0,1])}{\text{minimize}} \int_{\Theta} \lambda \rho(t) \frac{(p'(\theta))^2}{p(\theta)(1-p(\theta))} - \rho(\theta) u(\theta) p(\theta) d\theta \} \quad (P) \end{aligned}$$

where λ is a scaling parameter.

Simplification

- ▶ normalize the scaling parameter $\lambda = 1$
- ▶ normalize the state space $\Theta = [0, 1]$
- ▶ assume the prior ρ is uniform

Reparametrization

We consider the following change of variable:

$$p(\theta) = \frac{1+\sin(w(\theta))}{2}, p(\theta) \in [0, 1] \Leftrightarrow w(\theta) = \arcsin(2p(\theta)-1), w(\theta) \in [-\frac{\pi}{2}, \frac{\pi}{2}]$$

Our problem becomes ¹:

$$\underset{w \in AC(\Theta, [-\frac{\pi}{2}, \frac{\pi}{2}])}{\text{minimize}} \int_{\Theta} \lambda(w'(\theta))^2 - u(\theta) \frac{1 + \sin(w(\theta))}{2} d\theta. \quad (W)$$

We will apply tools from the **Calculus of Variations** to solve the above problem: existence, uniqueness, regularity and necessary conditions, .

¹To see this note that, $p'(\theta) = \frac{1}{2} \cos(w(\theta))w'(\theta)$ and $p(\theta)(1 - p(\theta)) = \frac{1}{4}(1 - \sin^2(w(\theta)))$

Existence

Theorem (Existence)

Suppose u is piecewise continuous, then Problem (P) admits a solution in $Lip([0, 1], [0, 1])$. Furthermore, if u is also single-crossing, then the optimal solution to Problem (P) belongs to one of the following two kinds:

- ▶ Pure boundary solution: $p(\theta) = 0$ for all $\theta \in [0, 1]$ or $p(\theta) = 1$ for all $\theta \in [0, 1]$
- ▶ Strict interior solution: $0 < p(\theta) < 1$ for all $\theta \in [0, 1]$

Uniqueness

Theorem (Uniqueness)

If the problem admits a non-constant interior optimal solution, then it is the unique solution.

Because the integrand $\frac{p'(\theta)^2}{p(\theta)(1-p(\theta))} - u(\theta)p(\theta)$ is convex in p . Two different continuous functions that differ in a neighborhood of positive measure imply that their convex combination attains a lower value.

Regularity

Theorem (Regularity)

Let p_* be a solution to Problem (P). If u is piecewise continuous, then p_* is twice differentiable almost everywhere. Furthermore, if $u \in C^k$ for some $k \in \mathbb{N}$, then $p_* \in C^{k+2}$.

As a result, the response curve exhibits no kinks or spikes.

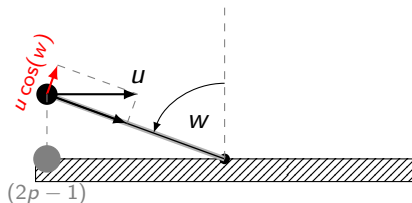
Characterizing the optimal (interior) solution

Theorem (Necessity)

Suppose u is piecewise continuous on $\bigcup_{i=0}^N (t_i, t_{i+1})$. Let w_* be a solution to Problem (W), then w_* satisfies the **integral Euler equation** together with **natural boundary conditions** and is continuously differentiable at each discontinuity of u :

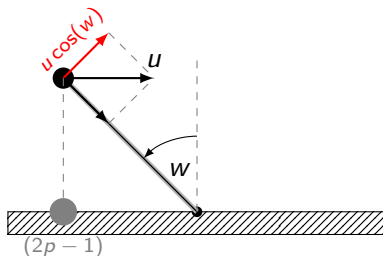
$$\begin{cases} w'_*(t) = \int_0^t -\frac{1}{4} u(s) \cos(w_*(s)) ds & \text{a.e. on } [0, 1] \\ w'_*(0) = w'_*(1) = 0 \end{cases} \quad (\text{NC})$$

Connection to a pendulum system



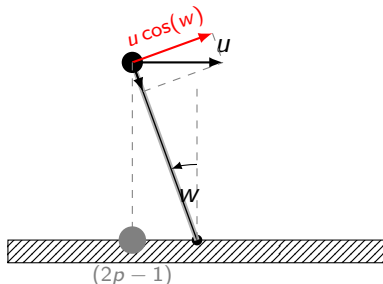
- ▶ $w(\theta)$: angle between rod and vertical axis. Angles to the left of the vertical axis are negative
- ▶ $u(\theta)$: magnitude of time-varying horizontal force. A force directed to the right is assigned a negative sign.
- ▶ Task: choose initial angle w_0 so that angular velocity is zero at $\theta = 0$ and $\theta = 1$
 $\Leftarrow w''(\theta) = -\frac{1}{4}u(\theta) \cos(w(\theta)) \quad w'(0) = w'(1) = 0$

Connection to a pendulum system



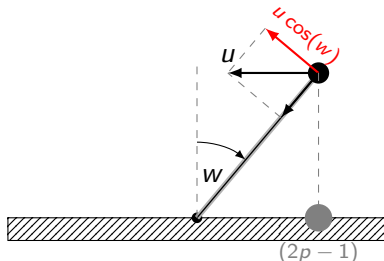
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Predicting Psychometric curve

Proposition (Simple conditions gives S-shaped curve)

If payoff u is single-crossing from below, then the optimal p_* is increasing.

Furthermore, suppose

- ▶ the prior is uniform
- ▶ the payoff u is increasing
- ▶ interior solution exists

then, the optimal solution p^* exhibits sigmoidal shape.

- ▶ The object first accelerates, then decelerates, and is always moving clockwise
- ▶ Its shadow accelerates at start and decelerates at the final

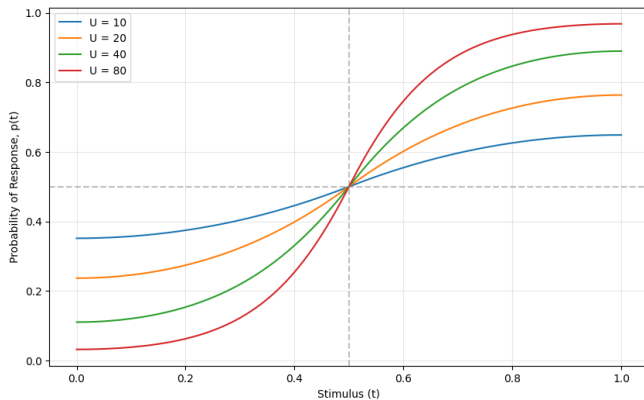
Comparative statics in payoffs

Proposition (stronger incentives responsiveness)

Suppose the prior is uniform, the payoff is anti-symmetric and single-crossing. If the incentive increases at every state, then:

- ▶ the response becomes more sensitive around the threshold $p'_*(\frac{1}{2}) \uparrow$
- ▶ the curve moves closer to correct responses $|p_*(\theta) - \frac{1}{2}| \uparrow$
- ▶ Stronger incentives implies stronger force, the object needs to start further away from the vertical line
- ▶ No overtake

Comparative statics in payoffs (illustration)



Comparative statics in prior

Proposition (Concentrated prior flattens the response)

Suppose the payoff is anti-symmetric and single-crossing, and the prior becomes more concentrated around the threshold in the sense that

$$\frac{\pi_2(\theta)}{\pi_1(\theta)} \text{ decreases as } |\theta - \tfrac{1}{2}| \text{ increases.}$$

. Then:

- ▶ the response becomes less sensitive around the threshold:
 $p'_{*,2}(\frac{1}{2}) \leq p'_{*,1}(\frac{1}{2})$;
- ▶ the curve moves toward indifference:
 $|p_{*,2}(\theta) - \frac{1}{2}| \leq |p_{*,1}(\theta) - \frac{1}{2}|$.

Flag: the model doesn't capture adaption to signal frequency.

Comparative statics in prior (illustration)

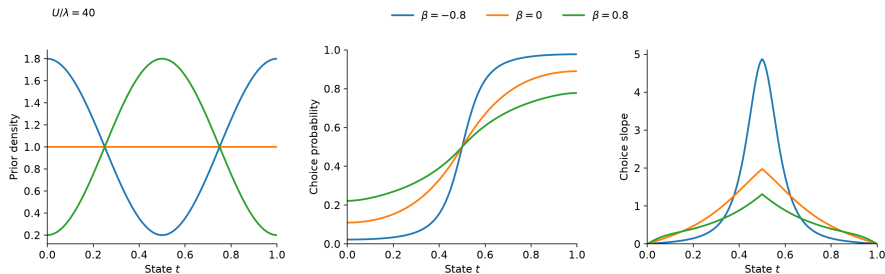
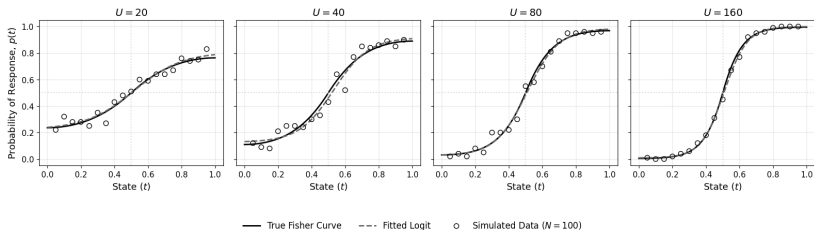


Figure: Concentration wrt MLR ordering

A simulation exercise

1. Derive the response (p^*) under the Fisher information cost and symmetric step payoff $u(\theta) = \begin{cases} -U, & \theta \in [0, \frac{1}{2}), \\ U, & \theta \in (\frac{1}{2}, 1], \end{cases}$
2. generate response counts from Binomial distribution $\text{Bin}(100, p^*(\theta))$ at different stimulus levels
3. Fit by maximum likelihood with standard psychometric function specification (logit): $p(\theta) = \gamma + (1 - \gamma - \kappa)F\left(\frac{\theta - \alpha}{\beta}\right)$



Flag: Fisher and standard logit are nearly observationally indistinguishable in lab setting.

Takeaway

- ▶ Fisher information cost provides a new explanation to sigmoidal response curve in perception-like economic decisions
- ▶ Connection between the optimal condition and a simple pendulum system
 - offer insight into the shape of the solution and comparative statics

Questions?

Thank you for your attention!

Any questions or comments?

References I

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- Bloedel, A. W. and Zhong, W. (2024). The cost of optimally-acquired information. *Working paper*.
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Change-of-Variable trick

It is without loss to work on the re-parametrized problem:

Lemma

The transformation $\Phi(p) := \arcsin(2p - 1)$ and $\Psi(w) := \frac{1+\sin(w)}{2}$ defines a bijection between the two classes of admissible solutions:

$$p = \frac{1 + \sin(w)}{2} \in \mathcal{D}^p \iff w = \arcsin(2p - 1) \in \mathcal{D}^w.$$

where

$$\mathcal{D}^p := \{p \in AC(\Theta, [0, 1]) : \int_{\Theta} \frac{(p'(\theta))^2}{p(\theta)(1-p(\theta))} d\theta < +\infty\}$$

$$\mathcal{D}^w := \{w \in AC(\Theta, [-\frac{\pi}{2}, \frac{\pi}{2}]) : \int_{\Theta} (w'(\theta))^2 d\theta < +\infty\}$$

Moreover, the value of the objective function is preserved under such a reparametrization, that is

$$\int_{\Theta} u(\theta)p(\theta) - \frac{p'(\theta)^2}{p(\theta)(1-p(\theta))} d\theta = \int_{\Theta} u(\theta) \frac{1+\sin(w(\theta))}{2} - (w'(\theta))^2 d\theta$$

if $p = \frac{1+\sin(w)}{2}$ and both are admissible.

Regularity

Theorem (Regularity)

Let p_* be a solution to Problem (P). If u is piecewise continuous, then p_* is twice differentiable almost everywhere. Furthermore, if $u \in C^k$ for some $k \in \mathbb{N}$, then $p_* \in C^{k+2}$.

As a result, the response curve exhibits no kinks or spikes.

Example

Consider the payoff $u = \begin{cases} -60 & \in [0, 0.35] \\ 30 & \in (0.35, 1] \end{cases}$, as can be seen from the numerical solution in Figure 2, $p_*''(t)$ has discontinuity at the point where u is discontinuous.

Regularity

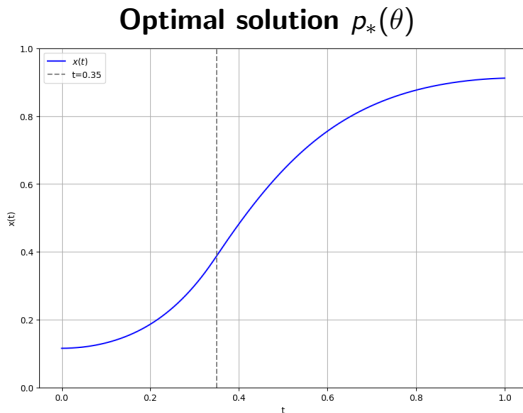


Figure: Example where $u \in \text{PWC}$ implies p_* twice differentiable almost everywhere

Regularity

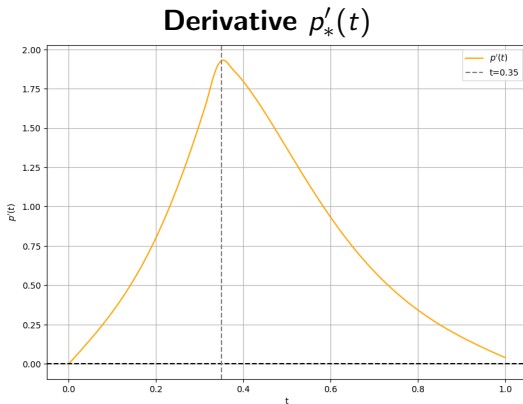


Figure: Example where $u \in \text{PWC}$ implies p_* twice differentiable almost everywhere

Regularity

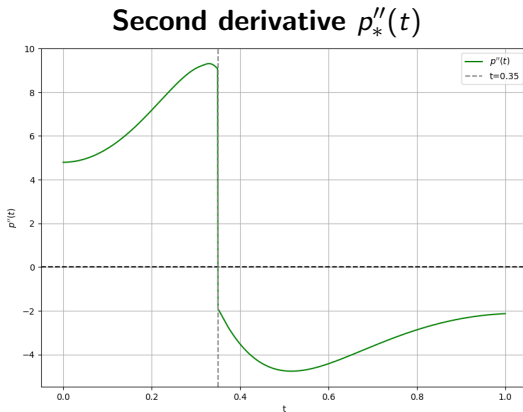


Figure: Example where $u \in \text{PWC}$ implies p_* twice differentiable almost everywhere

Predicting Psychometric curve

Proposition

Suppose the prior is uniform over Θ , the payoff is antisymmetric and single-crossing, i.e. $u(\theta) = -u(1 - t)$ on $\theta \in [0, 1]$ and $u(\theta) < 0$ on $[0, 1/2)$, then the optimal solution is interior. Let p_* be the interior maximizer of Problem (P). Then:

$$p_*''(t) > 0 \text{ on } [0, 1/2), \text{ and } p_*''(t) < 0 \text{ on } (1/2, 1].$$

2AFC

Example (2AFC)

Suppose the subject is rewarded for correct answers, then

$$u(\theta) = \begin{cases} -U, & \theta \in [0, 0.5), \\ U, & \theta \in (0.5, 1], \end{cases}$$

We have analytical solutions for the ODE of the simple pendulum:

$$w''(\theta) + \frac{U}{4} \cos(w(\theta)) = 0$$

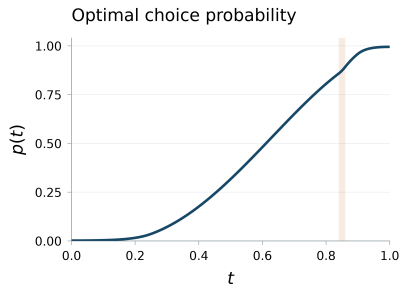
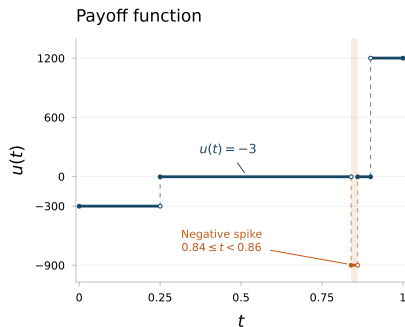
We just need to choose $w(0)$ and $w(1)$ to match the two curves:

$$w(0.5^-) = w(0.5^+) \quad w'(0.5^-) = w'(0.5^+)$$

Single-crossing is not enough for S-shape

The example: uniform prior, $\lambda = 1$,

$$u(\theta) = \begin{cases} -300, & 0 \leq \theta < 0.25, \\ -3, & 0.25 \leq \theta < 0.84, \\ -900, & 0.84 \leq \theta < 0.86, \\ -3, & 0.86 \leq \theta \leq 0.90, \\ 1200, & 0.90 < \theta \leq 1. \end{cases}$$



Negative payoff spike: $t \in [0.84, 0.86]$